GATE ME 90 Days Challenge

Day 53/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Engineering Materials	Materials & Heat Treatment	50 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on crystal structures and defects, phase rule and lever rule, the iron–carbon diagram and its microstructures, TTT diagrams, annealing, normalising, hardening, tempering and surface hardening, and the stress–strain behaviour of metals. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

For the face-centred cubic (FCC) crystal structure, the coordination number and the atomic packing factor are

(A) 8 and 0.68
(B) 12 and 0.74
(C) 12 and 0.68
(D) 6 and 0.52

Q2 NAT

Copper has an FCC structure with a lattice parameter of 0.3615 nm. Its atomic radius (in nm, round off to four decimal places) is _____

Q3 MCQ

At the eutectic point of a binary alloy system at constant (atmospheric) pressure, the number of degrees of freedom given by the Gibbs phase rule is

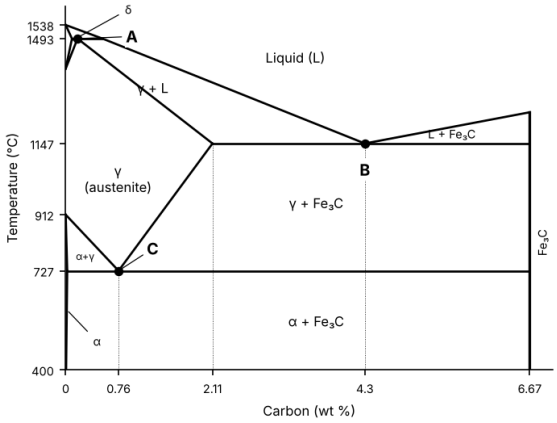
(A) 2
(B) 1
(C) 0
(D) 3

Q4 NAT

Taking the eutectoid composition as 0.76 % C, the carbon content of ferrite as 0.022 % and of cementite as 6.67 %, the mass percentage of ferrite in pearlite (round off to one decimal place) is _____

Q5 MCQ

Three invariant points A, B and C are marked on the iron–cementite diagram shown. The reaction that takes place on cooling through point B is



(A) δ + L → γ (peritectic)

(B) γ → α + Fe_3C (eutectoid)

(C) L → γ + δ

(D) L → γ + Fe_3C (eutectic, forming ledeburite)

Q6 NAT

A slowly cooled plain carbon steel is found to contain 30 % pearlite and 70 % proeutectoid ferrite at room temperature. Taking the eutectoid at 0.76 % C and ferrite at 0.022 % C, the carbon content of the steel (in %, round off to two decimal places) is _____

Q7 MCQ

Compared with a fully annealed specimen of the same hypoeutectoid steel, a normalised specimen has

(A) coarser pearlite and lower strength

(B) finer pearlite, a finer grain and higher strength and hardness

(C) a martensitic structure

(D) less pearlite and more ferrite with the same strength

Q8 NAT

The yield strength of a steel obeys the Hall–Petch relation $\sigma_y = \sigma_0 + k/\sqrt{d}$. It is 250 MPa for a grain size of 16 μm and 190 MPa for a grain size of 64 μm. The yield strength for a grain size of 36 μm (in MPa) is _____

Q9 MCQ

The maximum hardness that can be obtained in the martensite of a plain carbon steel depends mainly on

(A) the cooling rate, once it exceeds the critical cooling rate

(B) the austenite grain size

(C) the carbon content of the steel

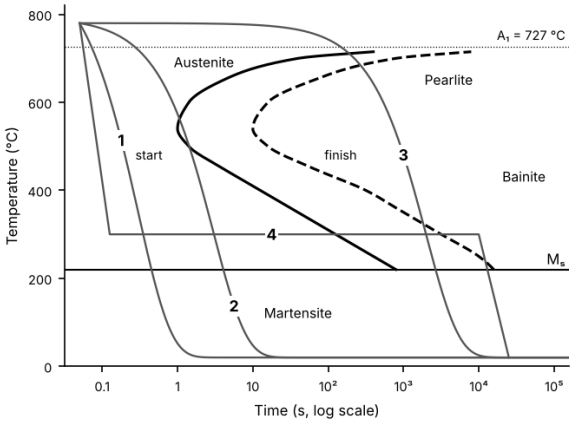
(D) the hardenability of the steel

Q10 NAT

A steel has a yield strength of 400 MPa and an elastic modulus of 200 GPa. Its modulus of resilience (in kJ/m³) is _____

Q11 MCQ

The TTT diagram of a eutectoid steel is shown with four cooling paths 1–4 (path 4 is held isothermally at 300 °C until transformation is complete). The path that gives a mixture of fine pearlite and martensite is



(A) path 1
(B) path 2
(C) path 3
(D) path 4

Q12 NAT

A cylindrical bar of 12.5 mm diameter and 50 mm gauge length is stretched uniformly (no necking) to a gauge length of 60 mm, the load then being 50 kN. Assuming constant volume, the true stress (in MPa, round off to one decimal place) is _____

Q13 MCQ

Which one of the following is a planar (two-dimensional) crystal defect?

(A) Vacancy
(B) Edge dislocation
(C) Interstitial atom
(D) Grain boundary

Q14 NAT

Aluminium is FCC with a lattice parameter of 0.405 nm. The interplanar spacing of its (111) planes (in nm, round off to four decimal places) is _____

Q15 MCQ

For a eutectoid steel, the microconstituents arranged in decreasing order of hardness are

(A) martensite, bainite, fine pearlite, coarse pearlite, spheroidite
(B) bainite, martensite, coarse pearlite, fine pearlite, spheroidite
(C) martensite, fine pearlite, bainite, spheroidite, coarse pearlite
(D) spheroidite, coarse pearlite, fine pearlite, bainite, martensite

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Answer Key

Q	Type	Answer
Q1	MCQ	(B) 12 and 0.74
Q2	NAT	0.1278 (accept 0.1276 to 0.1280)
Q3	MCQ	(C) 0
Q4	NAT	88.9 (accept 88.4 to 89.4)
Q5	MCQ	(D) $L \rightarrow \gamma + Fe_3C$ (eutectic, forming ledeburite)
Q6	NAT	0.24 (accept 0.23 to 0.25)
Q7	MCQ	(B) finer pearlite, a finer grain and higher strength and hardness
Q8	NAT	210
Q9	MCQ	(C) the carbon content of the steel
Q10	NAT	400
Q11	MCQ	(B) path 2
Q12	NAT	488.9 (accept 486.4 to 491.4)
Q13	MCQ	(D) Grain boundary
Q14	NAT	0.2338 (accept 0.2336 to 0.2340)
Q15	MCQ	(A) martensite, bainite, fine pearlite, coarse pearlite, spheroidite

Detailed Solutions

Q1 **Answer: (B) 12 and 0.74**

Concept: Coordination number = number of nearest neighbours; APF = volume of atoms in the cell/cell volume.

FCC has 4 atoms per cell with $a = 2\sqrt{2}\,r$: $APF = 4 \times (4/3)\pi r^3/(2\sqrt{2}\,r)^3 = \pi/(3\sqrt{2}) = 0.74$, and each atom touches 12 others.

So the answer is **(B) 12 and 0.74**. BCC: 8 and 0.68; simple cubic: 6 and 0.52; HCP (ideal): 12 and 0.74.

Exam tip: FCC and HCP are both close-packed (0.74); they differ only in the stacking sequence ABCABC… vs ABAB….

Q2 **Answer: 0.1278 (accept 0.1276 to 0.1280)**

Concept: In FCC the atoms touch along the face diagonal: $\sqrt{2}\,a = 4r$.

$r = \sqrt{2}\,a/4 = 1.4142 \times 0.3615/4$

$r =$ **0.1278 nm**

Exam tip: In BCC the atoms touch along the body diagonal instead: $\sqrt{3}\,a = 4r$.

Q3 **Answer: (C) 0**

Concept: Gibbs phase rule at constant pressure (condensed systems): F = C − P + 1.

At the eutectic point C = 2 and three phases (L, α, β) coexist: F = 2 − 3 + 1 = **0**, answer **(C)**.

The eutectic temperature and the three phase compositions are therefore fixed: the reaction is invariant, which is why the cooling curve shows a thermal arrest.

Exam tip: Eutectic, eutectoid and peritectic points are all invariant (F = 0) in a binary system.

Q4 **Answer: 88.9 (accept 88.4 to 89.4)**

Concept: Pearlite is the eutectoid mixture; apply the lever rule across the α + Fe_3C tie line at 0.76 % C.

Ferrite fraction = (6.67 − 0.76)/(6.67 − 0.022) = 5.91/6.648

Ferrite = **88.9 %**, cementite = 11.1 % (about 8 : 1).

Exam tip: Because ferrite plates are about 8 times as thick as cementite plates, pearlite looks like thin dark lines on a light background.

Q5 **Answer: (D) L → γ + Fe_3C (eutectic, forming ledeburite)**

Concept: The Fe–Fe_3C diagram has three invariant reactions: peritectic (1493 °C, 0.17 % C), eutectic (1147 °C, 4.3 % C) and eutectoid (727 °C, 0.76 % C).

Point B is at 4.3 % C and 1147 °C, where liquid freezes into austenite and cementite at once: L → γ + Fe_3C.

The eutectic mixture is called **ledeburite**: answer **(D)**. A is the peritectic and C the eutectoid (pearlite) point.

Exam tip: Eutectic: liquid → two solids; eutectoid: solid → two solids; peritectic: liquid + solid → another solid.

Q6 **Answer: 0.24 (accept 0.23 to 0.25)**

Concept: Lever rule just below 727 °C between ferrite (0.022 %) and pearlite (0.76 %).

Pearlite fraction = (C − 0.022)/(0.76 − 0.022) = 0.30

C = 0.022 + 0.30 × 0.738 = **0.24 %**

Exam tip: A quick estimate ignoring the carbon in ferrite is C ≈ 0.76 × (pearlite fraction) = 0.23 %.

Q7 **Answer: (B) finer pearlite, a finer grain and higher strength and hardness**

Concept: Annealing = furnace cooling from the austenite field; normalising = cooling in still air from about 40 °C above A_3 (or A_{cm}).

The faster air cooling lowers the transformation temperature, giving finer pearlite and a finer grain, and also slightly more pearlite (quasi-eutectoid).

Hence higher strength and hardness with good toughness: **(B)**.

Exam tip: Annealing → maximum softness and ductility; normalising → grain refinement and uniform structure (e.g. after forging or casting).

Q8 **Answer: 210**

Concept: Two data points fix σ_0 and k; then substitute the new grain size.

$1/\sqrt{(16\ \mu m)} = 250\ m^{-1/2}$, $1/\sqrt{(64\ \mu m)} = 125\ m^{-1/2}$: $k \times (250 - 125) = 60 \rightarrow k = 0.48$ MPa·√m, $\sigma_0 = 250 - 0.48 \times 250 = 130$ MPa

$1/\sqrt{(36\ \mu m)} = 166.7\ m^{-1/2}$: $\sigma_y = 130 + 0.48 \times 166.7 =$ **210 MPa**

Exam tip: Grain refinement is the only strengthening mechanism that also improves toughness.

Q9 **Answer: (C) the carbon content of the steel**

Concept: Martensite hardness comes from the carbon trapped in the BCT lattice.

Once the critical cooling rate is exceeded, cooling faster does not increase hardness; the hardness rises with carbon up to about 0.6–0.8 % C: **(C)**.

Alloying elements raise *hardenability* (depth of hardening), not the maximum hardness.

Exam tip: Hardness ≠ hardenability: a 0.4 % C alloy steel can harden deeper than a 0.8 % C plain steel, yet be less hard at the surface.

Q10 **Answer: 400**

Concept: Modulus of resilience = strain energy per unit volume up to the elastic limit = $\sigma_y^2/(2E)$.

$U = (400 \times 10^6)^2/(2 \times 200 \times 10^9) = 1.6 \times 10^{17}/4 \times 10^{11} = 4 \times 10^5$ J/m³

U = **400 kJ/m³**

Exam tip: Spring materials need a high σ_y^2/E; toughness (area under the whole curve) is a different property.

Q11 **Answer: (B) path 2**

Concept: A path that misses the start curve gives martensite; one that crosses both start and finish curves gives pearlite; one that crosses only the start curve transforms partly.

Path 2 passes through the nose region between the start and finish curves, so part of the austenite becomes fine pearlite and the rest turns to martensite below M_s: **(B)**.

Path 1 (fastest) → martensite; path 3 (slow) → coarse pearlite; path 4 → lower bainite (austempering).

Exam tip: The critical cooling rate is the slope of the line just touching the nose of the start curve.

Q12 **Answer: 488.9 (accept 486.4 to 491.4)**

Concept: In uniform plastic deformation the volume is constant: $A\ L = A_0\ L_0$, and true stress = load/current area.

$A_0 = \pi/4 \times 12.5^2 = 122.72$ mm²; $A = 122.72 \times 50/60 = 102.27$ mm²

$\sigma_{true} = 50\ 000/102.27 =$ **488.9 MPa** (engineering stress 407.4 MPa; $\sigma_{true} = \sigma_{eng}(1 + e) = 407.4 \times 1.2$).

Exam tip: The true strain here is $\ln(60/50) = 0.182$, smaller than the engineering strain 0.2.

Q13 **Answer: (D) Grain boundary**

Concept: Defects are classified by dimension.

Point (0-D): vacancies, interstitial and substitutional atoms. Line (1-D): edge and screw dislocations. Planar (2-D): grain boundaries, twin boundaries, stacking faults, free surfaces. Volume (3-D): voids, inclusions.

Answer: **(D) grain boundary**.

Exam tip: Plastic deformation in metals takes place by the motion of dislocations (line defects).

Q14 **Answer: 0.2338 (accept 0.2336 to 0.2340)**

Concept: For cubic crystals $d_{hkl} = a/\sqrt{(h^2 + k^2 + l^2)}$.

$d_{111} = 0.405/\sqrt{3} = 0.405/1.7321$

$d_{111} =$ **0.2338 nm**

Exam tip: The (111) planes are the most widely spaced (close-packed) planes in FCC, which is why slip occurs on them.

Q15 **Answer: (A) martensite, bainite, fine pearlite, coarse pearlite, spheroidite**

Concept: Hardness rises as the carbide becomes finer and as carbon is trapped in the iron lattice.

Martensite (supersaturated BCT) is the hardest; bainite has very fine carbides; fine pearlite has thinner lamellae than coarse pearlite; spheroidite (globular carbides) is the softest.

Answer: **(A)**. Ductility follows the reverse order.

Exam tip: Spheroidising is used on high-carbon steels to give the best machinability and cold formability.

What next?

Day 53 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 54/90: Week 9 Revision. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.